

Supplementary Online Content

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eReferences

This supplementary material has been provided by the authors to give readers additional information about their work.

eMethods. Supplemental Methods

1. Description of the study cohort

1.1. Study population

The Coronary Atherosclerosis Disease Early Identification and Risk Stratification by Noninvasive Imaging (CREATION, clinical trial registration number: NCT03518437) study is a large and newly developed cohort study with participants selected from outpatients nationwide primarily from north, southwest and southeast of China. This cohort was organized by Beijing Fuwai hospital affiliated to Chinese Academy of Medical Sciences and National Center for Cardiovascular Diseases of China, with major focus on identifying risk factors that lead to coronary artery atherosclerotic plaque progression and events of coronary artery diseases (CAD). Fuwai Hospital is the largest hospital in China, specializing cardiovascular diseases treatment, prevention and research. Nowadays, more than 6 million outpatients, from 32 provinces in the nation and 6 countries worldwide have received their cares.

The CREATION cohort recruited outpatients of adults (aged>25) that visit Fuwai hospital who are suspected of having CAD without history of coronary revascularization other heart disease and who underwent cardiac computed tomography between 2015 and 2017. Similar study design could be found elsewhere¹. The participants had relatively low pretest probability of CHD (27.3±15.1%), with asymptomatic patients (n=2587), non-anginal chest pain (n=5721), atypical angina (n=449), typical angina (n=110), therefore, this cohort can represent the low risk natural population. Typical angina was defined as having all of the following criteria: (1) substernal chest pain or discomfort that was (2) provoked by exertion or emotional stress and (3) relieved by rest and/or nitroglycerine. Symptoms were classified as atypical angina when only two of these criteria were present, and as nonanginal chest pain if one or none of these characteristics were reported.

This study built air pollution and traffic exposures upon the CREATION cohort participants who had accurate geocoded addresses for exposure assessment at baseline. All the participants will be followed-up for CAD disease events collection and some of them will be re-examined for CT scans after 3 and 5 years of baseline survey. In addition to clinical information collected during the hospital visit and subsequent visits for follow-up treatment or physical examinations, an extensive set of cardiovascular biomarkers of was collected in this cohort. Furthermore, a subset of the participants completed detailed survey on environmental risk factors, including building environment, socio-economic status and residential mobility. The large cohort size and rich subclinical and biological database make it feasible to associate long-term air pollution and traffic exposures with outcomes that may define pathways by which air pollution contributes to the atherosclerotic cardiovascular disease. Detailed participants selection and spatial distributions of their residences exposed to air pollution were shown as below.

1.2. Clinical data

A structured interview was conducted before CT exam to collect information on demographic characteristics, the presence and duration of cardiovascular risk factors and biochemical results. Hypertension was defined as a previously established diagnosis, systolic blood pressure ≥ 140 mmHg, diastolic blood pressure ≥ 90 mmHg, or antihypertensive medication use. Hypercholesterolemia was defined according to the National Cholesterol Education Panel (NCEP) guidelines or by the current use of lipid-lowering medication. Diabetes mellitus was defined as a previously established diagnosis, insulin or oral hypoglycemic therapy. Current smoking was defined as any cigarette smoking within 1 year prior to the CT exam. Alcohol consumption was defined as drinking more than twice a week for more than a year. Meat consumption was defined as eating meat more than three times a week, more than 300 grams for each time. Physical activity was defined as frequency of aerobic exercise for more than an hour. Coronary heart disease was defined as at least 1 coronary segment with a lesion of $\geq 50\%$ luminal stenosis in diameter. Any disagreements between two readers were solved by consensus. The duration of the risk factors was defined as the time between made a definite

diagnosis and underwent CT. The results of the biochemical test were performed within 1 month of the CT exam.

2. Estimating historical PM_{2.5}

To extrapolate PM_{2.5} concentrations back in time, we leveraged a large dataset with annual average satellite-derived PM_{2.5} estimates (10x10 km²) from 2004 to 2014². We converted our fine-scale point-based PM_{2.5} predictions in 2014 to the historical years using the ratio of the satellite-derived PM_{2.5} predictions between 2014 and the target year (see equation 1). This calculation is based on an assumption that, for each point-based prediction from our fine-scale model, the spatial contrast of PM_{2.5} is constant between the two years within a 10x10km grid cell, but varies between grid cells. We then calculated the long-term PM_{2.5} concentrations averaged back for 3-, 4-, 5-, 6-, and 10-year respectively, ahead of baseline since 2015, to assess the effect of different lengths of exposure on difference of CAC.

$$PM2.5_{year(i)}^{LUR(point)} = PM2.5_{2014}^{LUR(point)} \times \frac{PM2.5_{year(i)}^{SAT(10km)}}{PM2.5_{2014}^{SAT(10km)}} \quad (\text{Equation 1})$$

Where year (i) donates the target year to be extrapolated to between 2004 and 2014, SAT donates the 10x10 km² annual averaged satellite-derived PM_{2.5} estimates, and LUR donates our point-based hierarchical land use regression modeling predictions.

eTable 1 Summary Statistics and Correlations of the Exposure Variables in the Study Participants

Exposure	Min	Q1	Median	Q3	Max	Correlation coefficients			
						PM _{2.5}	NO ₂	O ₃	Distance to roadway (feet)
PM _{2.5} (µg/m ³)	17.5	52.9	80.4	85.5	108.0	1	0.72	0.53	-0.1
NO ₂ (µg/m ³)	8.2	29.4	42.4	54.5	62.8	0.72	1	0.56	-0.24
O ₃ (µg/m ³)	49.9	88.9	98.7	101.5	122.1	0.53	0.56	1	-0.14
Distance to roadway (feet)	12.1	98.4	218.8	134.3	4955.3	-0.1	-0.24	-0.14	1

eTable 2 Main and Sensitivity Analyses for Percent Change and 95% CI in CAC Associated With a Difference in Long-term Exposure Variables^a Among the CREATION Participants.

ID	N	Staged models ^b	PM _{2.5} (30 µg/m ³)	NO ₂ (20 µg/ m ³)	O ₃ (15 µg/ m ³)	Distance to road (reduction by 50%)
1	8168	Model 1	16.6 (8.4, 25.5)	8.5 (1.2, 16.3)	9.4 (2.5, 16.7)	2.5 (-0.1 ,5.3)
2	8168	Model 1 (+risk factor)	17.4 (8.9, 26.5)	11 (3.1, 19.4)	8.9 (1.9, 16.4)	3.0 (0.3 ,5.8)
3	8168	Model 2 (+area-level covariates, primary model)	29.6 (15.7, 45.2)	33.2 (16.4, 52.4)	10.8 (0.9, 21.8)	3.2 (0.3 ,6.2)
4	8168	Model 3 (+medications)	27.6 (13.9, 42.8)	31.4 (14.9, 50.2)	9.5 (-0.2, 20.2)	3.0 (0.1 ,6.0)
5	6538	Model 3 (+biomarkers)	28.8 (13.8, 45.8)	31.5 (13.5, 52.3)	11.5 (0.8, 23.4)	3.1 (0.2 ,6.1)
6	8168	Model 3 (+multi- pollutants)	27.2 (10.8, 46.1)	24.5 (3.6, 49.7)	9.0 (-1.4, 20.4)	2.4 (-0.6 ,5.4)
7	8168	Model 3(spatial cluster)	30.2 (16.1, 46.0)	33.0 (16.2, 52.2)	10.5 (0.5, 21.4)	3.1 (0.2 ,6.1)
8	8015	Model 3 (excluding extreme values)	47.8 (33.0, 64.2)	57.4 (39.0, 78.3)	11.9 (2.6, 22.2)	3.2 (0.3 ,6.2)
9	8168	Model 3 (two-year average exposure)	27.4 (14.8, 41.4)	26.8 (10.8, 45.2)	9.5 (0.3, 19.5)	
10	PM _{2.5} : 440 NO ₂ : 3617 O ₃ : 4468	Model 3 (below national standard)	22.8 (-9.1, 66.0)	24.0 (6.4, 44.5)	7.4 (-0.5, 16.1)	
11	1732	Model 3 (postmenopausal women)	34.5 (5.8, 70.9)	26.6 (-2.8, 65.0)	4.1 (-8.5, 18.5)	

^aIncrement difference of each exposure variables for the health effect estimates are 30 µg/m³ for PM_{2.5}, 20 µg/m³ for NO₂, 15 ug/m³ for O₃ and 50% decrease for road distance;

^bModel 1= age, gender;

Model 2= model1+BMI, smoking status, smoking years, cigarettes per day, alcohol consumptions, education, exercise;

Model 3= model2+ urbanization, regions, distance to hospital, local Beijing (yes/no);

Model 4=model 3+ antihypertensive medicine and use of statin;

Model 5=model 3+biomarkers: total cholesterol, high and low density lipoprotein, triglycerides, and high-sensitivity C-reactive protein

Model 6=model 3+multi-pollutants: PM_{2.5}, NO₂, O₃, distance to roadway were modeled simultaneously.

Model 7=model 3+random slope accounting for county-level area effects;

Model 8=model 3 without extreme observations identified by cook's distance values;

Model 9=model 3 with exposure to air pollutants averaged between 2014 and 2015.

Model 10=model 3 with exposure below the national standard (i.e. $<35 \mu\text{g}/\text{m}^3$ national standard for annual average $\text{PM}_{2.5}$, $<40 \mu\text{g}/\text{m}^3$ national standard for annual average NO_2 , and $<100 \mu\text{g}/\text{m}^3$ US standard for annual average O_3); increment difference of exposure for the health effect estimates are $10 \mu\text{g}/\text{m}^3$ for both $\text{PM}_{2.5}$ and NO_2 .

Model 11=model 3 restricted to postmenopausal women.

eTable 3 Sensitivity Analyses With Detailed Individual-Level Risk Factors for Associations Between Long-term Air Pollution Exposure Variables^a and CAC-score Among a Subset of Participants (n = 1850)

Staged models	N	PM _{2.5} ^b (30 µg/ m ³)	NO ₂ (20 µg/ m ³)	O ₃ (15 µg/ m ³)	Distance to road (reduction by 50%)
Model 2 ^b	1850	36.5 (14.0 ,63.4)	36.9 (8.6 ,72.6)	18.7 (1.2 ,39.1)	6.1 (-1.2 ,14.0)
Model 2+Air conditioner	1850	40.5 (17.1 ,68.6)	43.1 (13.0 ,81.2)	19.2 (1.4 ,40.1)	5.7 (-1.7 ,13.5)
Model 2+Heating	1850	39.1 (16.1 ,66.6)	41.4 (12.0 ,78.5)	18.5 (1.0 ,39.0)	6.0 (-1.4 ,13.8)
Model 2+Air purifier	1850	39.1 (16.1 ,66.6)	41.3 (11.9 ,78.4)	18.5 (1.0 ,39.0)	5.7 (-1.6 ,13.6)
Model 2+Cooking ^c	1850	38.6 (15.7 ,66)	40.9 (11.6 ,77.8)	17.5 (0.1 ,37.8)	5.5 (-1.8 ,13.4)
Model 2+Ventilation ^d	1850	37.9 (15.0 ,65.3)	40.7 (11.4 ,77.7)	18.8 (1.2 ,39.5)	5.4 (-1.9 ,13.3)
Model 2+Environmental tobacco smoking	1850	36.5 (14.1 ,63.4)	36.8 (8.5 ,72.5)	18.7 (1.2 ,39.1)	6.2 (-1.1 ,14.1)
Model 2+Employment	1850	35.8 (13.4 ,62.5)	36.3 (8.1 ,72.0)	18.3 (0.9 ,38.7)	6.2 (-1.1 ,14.1)
Model 2+Income ^e	1850	36.0 (13.6 ,62.7)	36.8 (8.5 ,72.4)	18.3 (1.0 ,38.6)	5.7 (-1.6 ,13.5)
Model 2+Pet at home	1850	36.5 (14.0 ,63.5)	36.9 (8.6 ,72.8)	18.7 (1.2 ,39.1)	6.1 (-1.2 ,13.9)
Model 2 (no moving history)	1762	36.0 (13.1 ,63.6)	37.9 (8.8 ,75.0)	16.1 (-1.5 ,36.9)	5.6 (-1.9 ,13.7)

^aIncrement difference of each exposure variables for the health effect estimates are 30 ug/m³ for PM_{2.5}, 20 ug/m³ for NO₂, 15 ug/m³ for O₃, and 50% decrease for road distance;

^bModel adjusted for age, gender, BMI, smoking status, smoking years, cigarettes per day, alcohol consumptions, education, exercise, urbanization, regions, distance to hospital, local Beijing (yes/no);

^cCooking includes self-cooking (yes/no), dinner at home (yes/no), and cooking stove (electricity/gas);

^dVentilation includes use of fans for cooking (yes/no) and frequency;

^eIncome per month is categorized by (<450, 450-750, 750-1200, 1200-2250,2250-3000, >3000 US dollars)

eTable 4 Effect Modifications in Associations of CAC With Long-term Exposure to PM_{2.5} and NO₂ for Selected Demographics and Risk Factors

Percent Change in CAC (95% CI) ^a	PM _{2.5} (µg/m ³) ^b	P-value	NO ₂ (µg/m ³) ^b	P-value
Gender		0.01		<0.01
Female	17.6 (2.6, 34.8)		17.9 (1.4 ,37.2)	
Male	42.2 (24.3, 62.7)		45.7 (25.3 ,69.5)	
Age		<0.01		<0.01
< 60	17.1 (3.0, 33.2)		21.0 (4.6 ,40.0)	
≥ 60	50.1 (28.8, 75.0)		55.5 (31.8 ,83.6)	
BMI		0.22		0.23
< 30	31.3 (17.0, 47.4)		32.7 (15.9 ,51.8)	
≥ 30	10.3 (-16.6, 45.9)		14.4 (-11.9 ,48.5)	
Smoking status		0.46		0.45
Never, Former	26.7 (12.2, 43.2)		30.2 (13.1 ,49.8)	
Current	34.6 (14.8, 57.8)		33.8 (12.6 ,58.8)	
Diabetes mellitus		0.01		0.09
No	27.0 (10.8, 45.5)		16.7 (8.0, 26.1)	
Yes	62.2 (30.9, 101.0)		31.2 (13.9, 51.0)	
Regions		0.38		0.02
North	31.5 (15.5, 49.6)		42.0 (21.7 ,65.6)	
Southeast	15.1 (-10.4, 47.8)		-9.2 (-32.1 ,21.4)	
Southwest	72.3 (-2.7, 205.2)		48.9 (-3.5 ,129.7)	
Urbanization		0.20		0.82
Urban	22.8 (6.6, 41.3)		30.0 (10.9 ,52.3)	
Rural	37.5 (18.9, 59.0)		33.1 (10.5 ,60.2)	
Statin		0.74		0.86
No	17.1 (8.2, 26.8)		30.5 (13.6 ,50.0)	
Yes	2.3 (0.8, 3.7)		28.7 (7.5 ,54.0)	

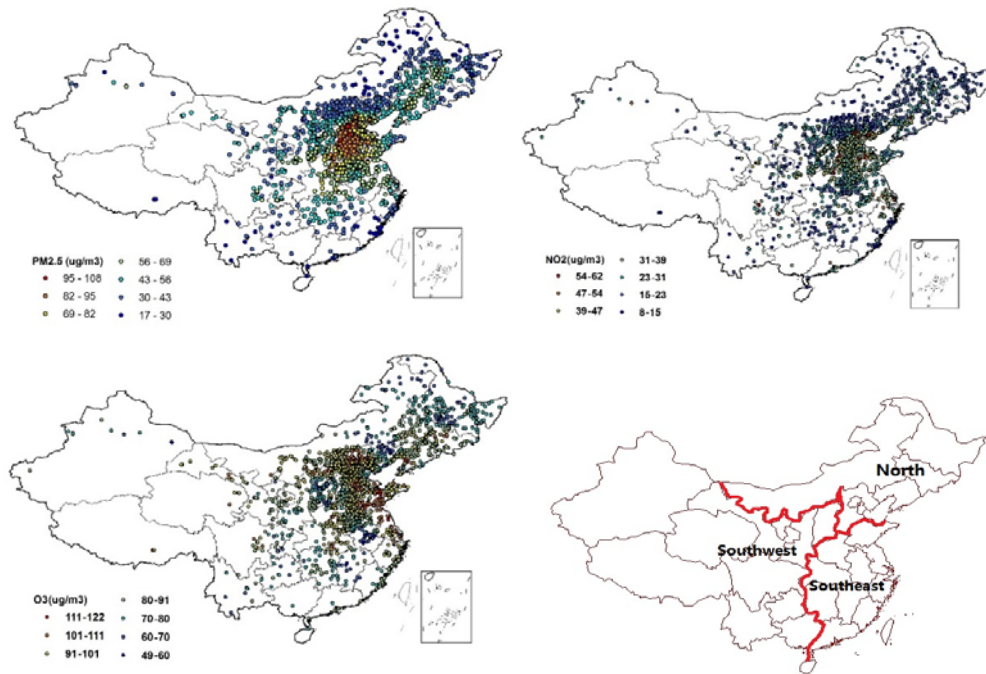
^aModel adjusted for age, gender, BMI, smoking status, smoking years, cigarettes per day, alcohol consumptions, education, exercise, urbanization, regions, distance to hospital, local Beijing (yes/no);

^bIncrement difference of each exposure variables for the health effect estimates are 30 ug/m³ for PM_{2.5}, 20 ug/m³ for NO₂;

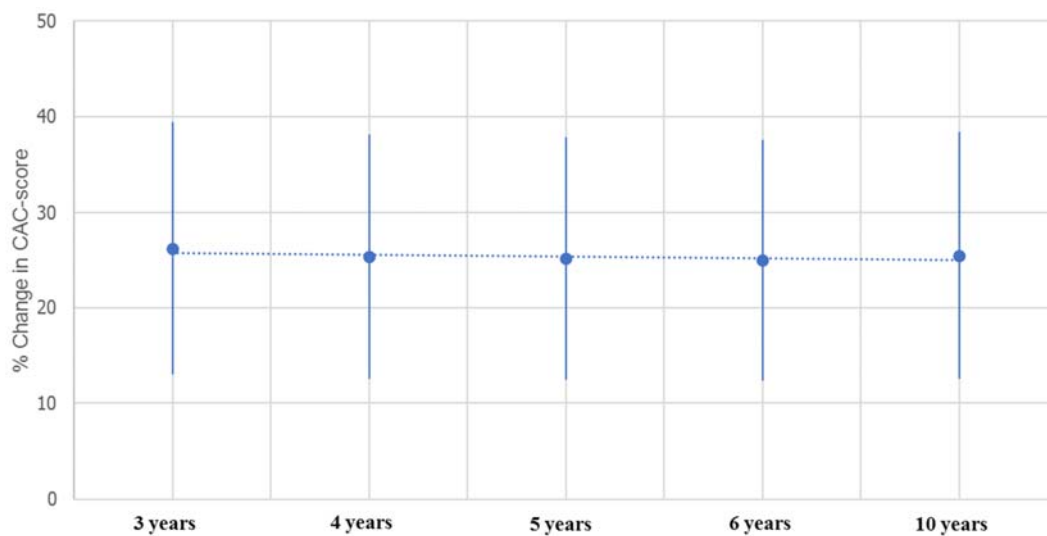
eTable 5 Comparisons of Health Effect Estimates in and Outside Beijing

%Change in CAC (95%CI) ^a	In Beijing (n=2856)	Outside Beijing (n=5312)	p-value
PM _{2.5}	10.3 (1.8 ,19.4)	16.1 (9.4 ,23.1)	0.06
NO ₂	9.2 (0.9 ,18.3)	14 (7.1 ,21.4)	0.25
O ₃	-3.2 (-11.5 ,6)	9.9 (3.5 ,16.7)	0.43

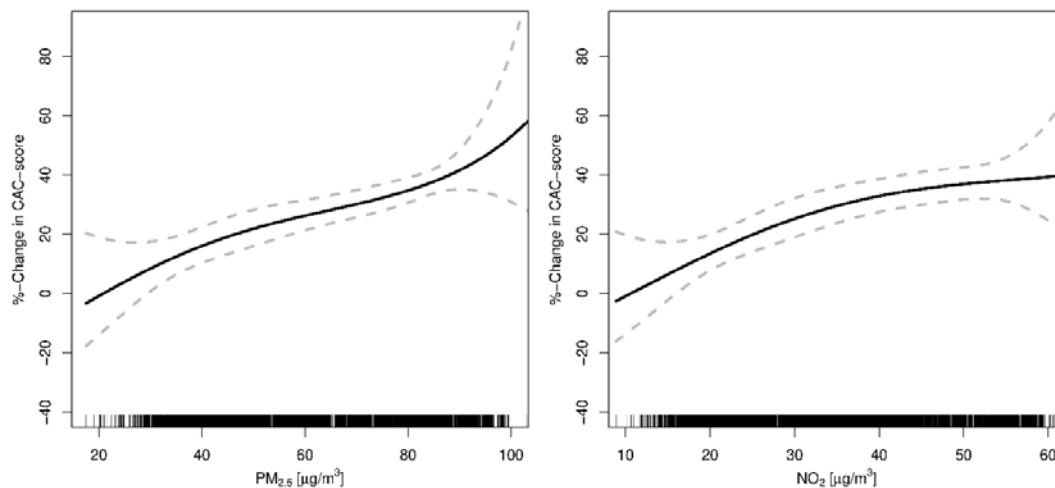
^aModel adjusted for age, gender, BMI, smoking status, smoking years, cigarettes per day, alcohol consumptions, education, exercise, urbanization, regions, distance to hospital; Health effect estimates were assessed by a SD increase of exposure (Table S4) in or outside Beijing.



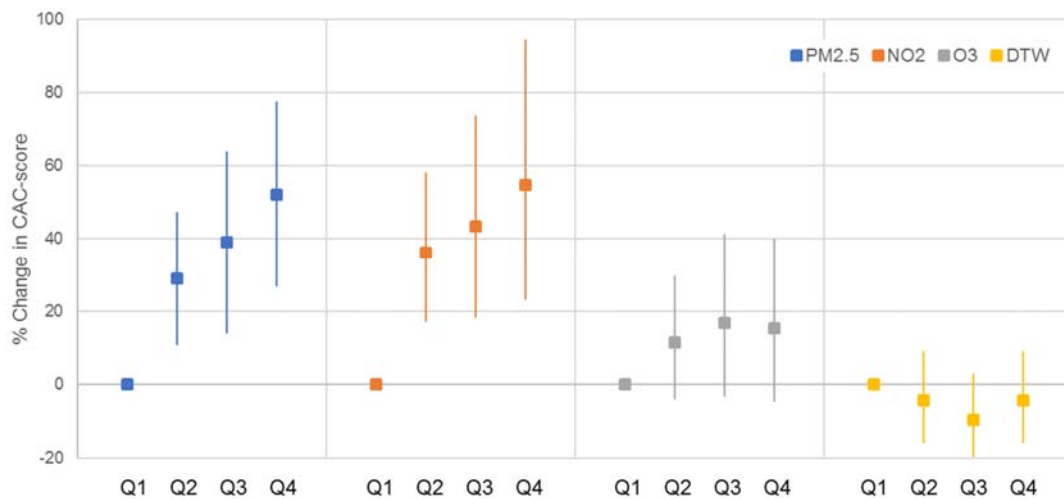
eFigure 1 Spatial Distribution of Estimated Annual Mean Exposure to PM_{2.5}, NO₂, and O₃ in the Year of 2015 at the Participants' Home and Regional Classifications.



eFigure 2 Percent Change and 95% CI in CAC Associated With PM_{2.5} (30 µg/m³) Averaged Over Multiple Years Prior to Baseline Using Land Use Regression Model With Adjustment for Satellite-derived PM_{2.5} Observations. Detailed Information on Exposure Estimation for PM_{2.5} Is Described in eMethod: Long-term PM_{2.5} Predictions.

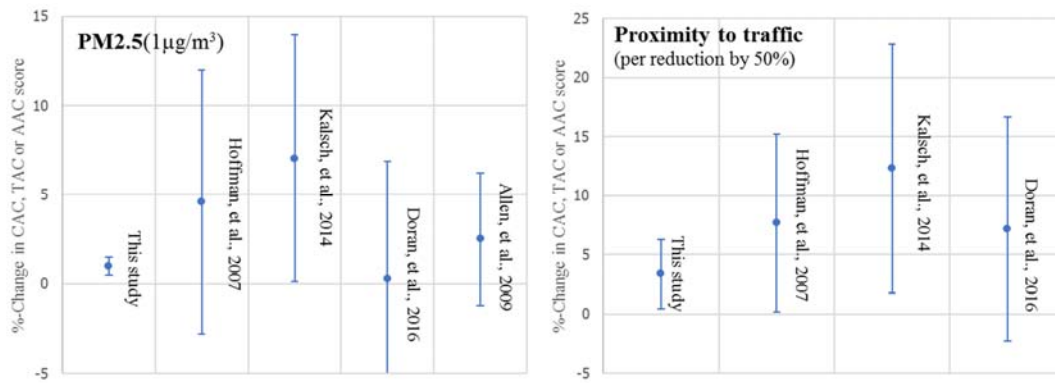


eFigure 3 Concentration-Response Relationships Between Long-term Exposure to PM_{2.5} and NO₂ and the Estimated Percent Change in CAC Score, Modeled by a Natural Spline With 3 Degrees of Freedom.



eFigure 4 Associations Between CAC and Long-term Exposure Variables Categorized by Quartiles

(p-Values for Trends of Categorical Exposure Are <0.001 for PM_{2.5} and NO₂, 0.382 for O₃ and 0.237 for DTW). DTW: Distance to roadway. Q1 is the reference exposure.



eFigure 5 Comparisons of Heath Effect Estimates for Associations of PM_{2.5} and Proximity to Traffic Exposures With Degrees of CAC, Thoracic Aortic Calcification (TAC) or Abdominal Aortic Calcification (AAC) Between This Study and the Previous Studies³⁻⁶.

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